Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name:** JAAAR (Just Another Agentic AI Researcher)
b. **Team leader name:** Mir Jasimuddin
c. **Problem Statement:** Future of Work & Enterprise Productivity - turning process knowledge that lives in people's heads into governed, trustworthy automation

Brief about the idea

Understudy - show it once, get automation you can trust

Every enterprise runs on repetitive screen work - invoices, onboarding, data entry - and the know-how lives in people's heads. Automating one process today takes a specialist, weeks of scripting and a PDF of rules nobody enforces.

Understudy lets anyone **record a task once**. Gemini on Vertex AI watches the recording, turns the written SOP into **rules that block bad data**, writes and proves the automation code, runs it for real, heals it when the app changes - and ships it only after a **verified run**.

IN

Screen recording of the task (with narration)
Address of any web app
The SOP in plain language - any language

OUT

A tested automation suite (Playwright, Selenium, ... or a Gemini agent)
The SOP rules inside the script
Evidence of a real, verified run - as one ZIP



Five gated steps: a step cannot start until the previous one passes, and the download is refused until all five do. **Gemini understands; deterministic checks decide.**

Opportunities

How is it different?

- RPA recorders replay brittle click paths; free-roaming AI agents act without guardrails or proof
- Understudy learns from video + narration + the live page + the SOP
- It ships the business rules inside every generated script
- It proves success in the target system's own ledger, not by an exit code

How does it solve the problem?

- Watch: Gemini rebuilds every step, time-stamped to the video, checked on the live page
- Govern: the SOP becomes confirmed rules; a rule-breaking value is blocked before anything runs
- Build + Execute: proven locators, restricted network, self-healing with an approved diff
- Export: only a verified run is packaged

USP

- Video to governed code in minutes, for people who have never written automation
- Automation Difficulty Score picks a generated script or a run-time Gemini agent
- Heals, then remembers: lessons become learned skills that work even offline
- Model Armor on every input; every AI call visible to the admin

List of features offered by the solution

Multimodal Watch

Gemini reads video frames + narration; each step links to its moment in the video

Live-page grounding

Every selector is verified on fresh loads of the real page before any run

Difficulty Score

Ten parameters Gemini picks for this process; routes traditional vs agentic

SOP -> rules

Plain-language SOP becomes rules the user confirms, each quoting its source sentence

Compliance gate

A rule-breaking payload is blocked and can be fixed in place; rules ship in the ZIP

Any tool, or an agent

Playwright, Selenium, Cypress, Robot, REST Assured ... or the Understudy Gemini Agent

Self-healing

Gemini diagnoses a failed run, shows the fix as a diff, re-runs after approval, learns

Restricted execution

Only the target is reachable; no keys in the run; time and memory limits

Evidence-backed export

ZIP with code, payload, rules, run report and CI-ready JUnit output

Model Armor

Layered injection defence on every input, including text read from videos and pages

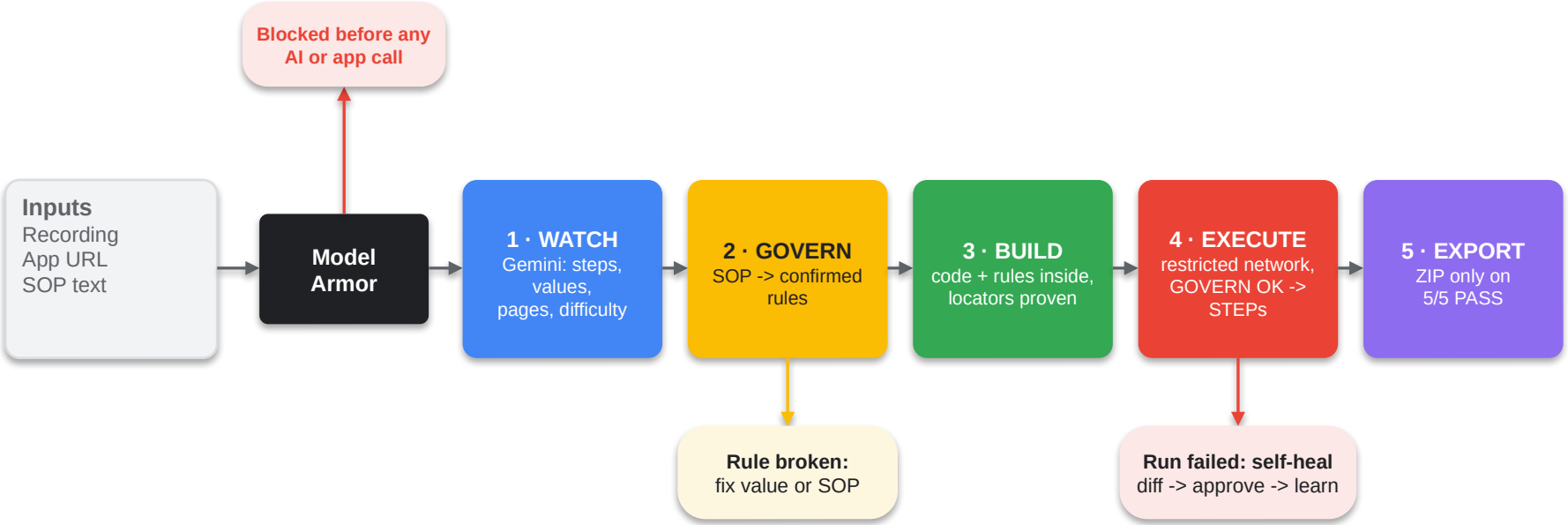
Mission Control

Admin console: telemetry of every AI call, allowed tools/sites/versions, users

Learned skills

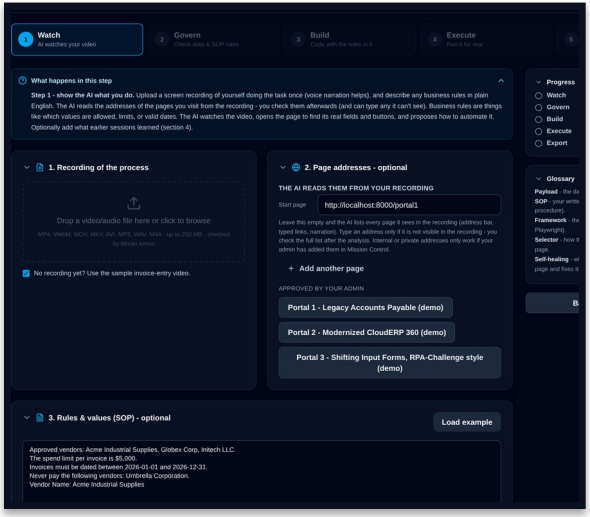
Validated AI answers are reused - keeps working offline or out of quota

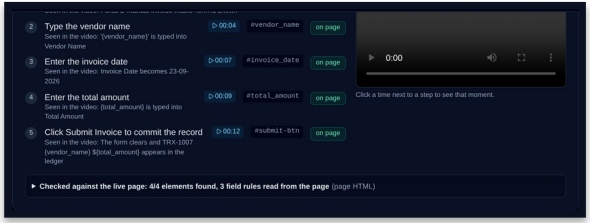
Process flow diagram or Use-case diagram

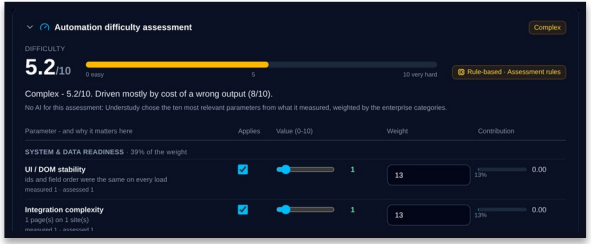


Use case: an accounts-payable clerk records entering one supplier invoice. The SOP's "$5,000 spend limit" becomes a rule; a mistyped $7,500 is blocked in Govern; the corrected run enters the invoice, and the record is verified in the portal's ledger (TRX-1002, $450.00).

Wireframes/Mock diagrams of the proposed solution (optional)





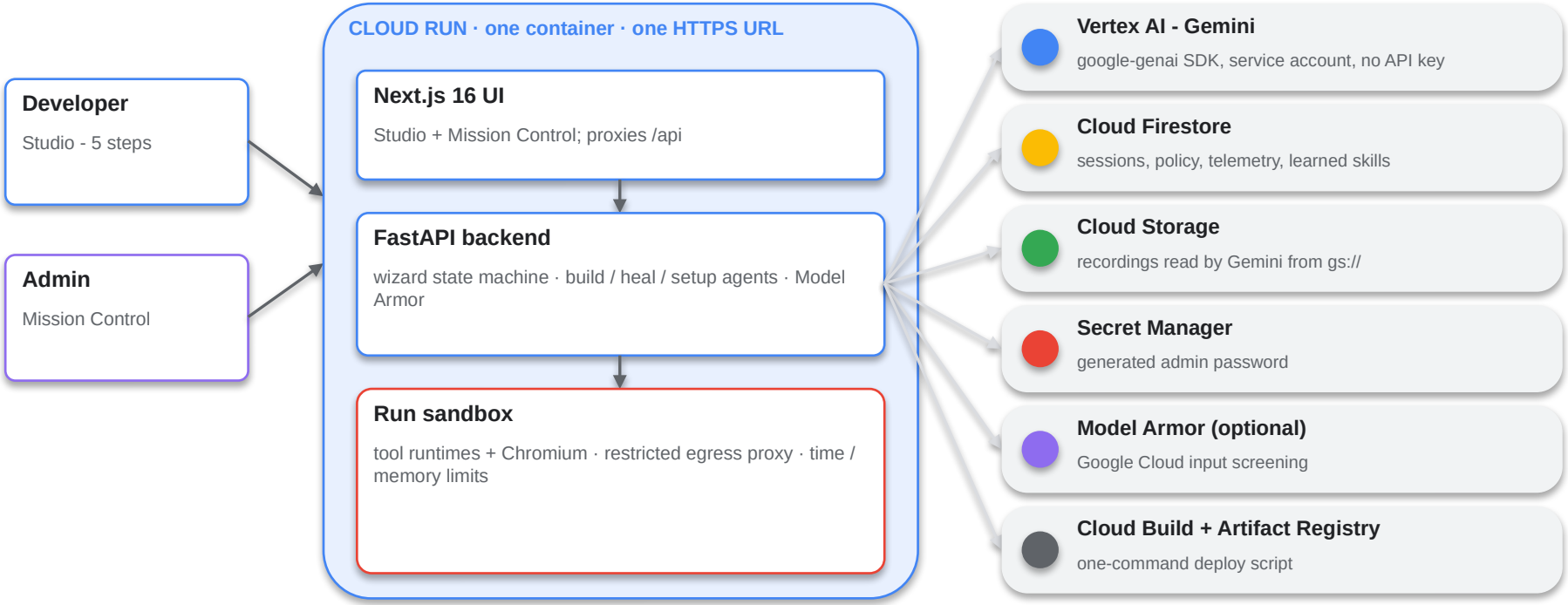


Step 1 inputs: recording, approved pages, SOP

Watch: time-stamped steps checked on the live page

Automation Difficulty Score and routing

Architecture diagram of the proposed solution



Target web app is reached only from the run sandbox. Least-privilege service account; state shared across instances via Firestore; deploy_cloudrun.sh builds, deploys and checks the live URL.

Technologies to be used in the solution

GOOGLE AI

- Gemini via the Gen AI SDK (google-genai)
- Vertex AI (keyless, ADC) or AI Studio
- Multimodal video understanding
- 28 registered prompts, schema-validated

GOOGLE CLOUD

- Cloud Run (single container)
- Cloud Firestore · Cloud Storage
- Secret Manager · Cloud Build · Artifact Registry
- Model Armor (optional stage)

APPLICATION

- Python 3 · FastAPI · Uvicorn
- Next.js 16 · React 19 · Tailwind 4
- Playwright + Chromium (probe, agent)
- pytest - 855 tests · GitHub Actions CI

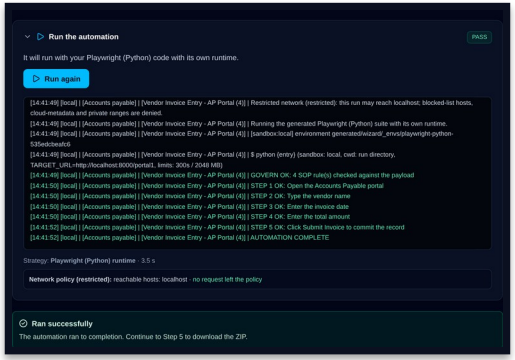
AUTOMATION TOOLS GENERATED

- Playwright (Python / JS) · Puppeteer
- Selenium · Cypress · WebdriverIO
- Robot Framework · REST Assured
- Understudy Gemini Agent (agentic)

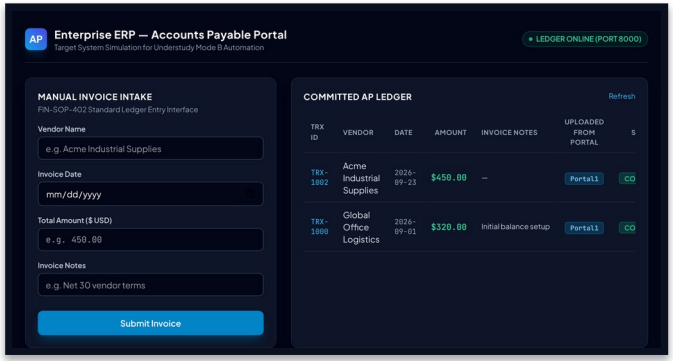
Snapshots of the prototype



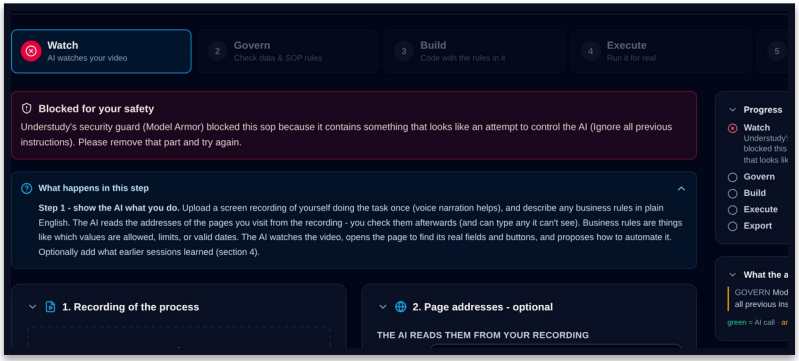
Govern blocks a $7,500 invoice (SOP limit $5,000)



Execute: GOVERN OK -> STEP 1-5 OK on a restricted network



Proof: the record is in the target app's ledger



Model Armor blocks a prompt injection hidden in the SOP

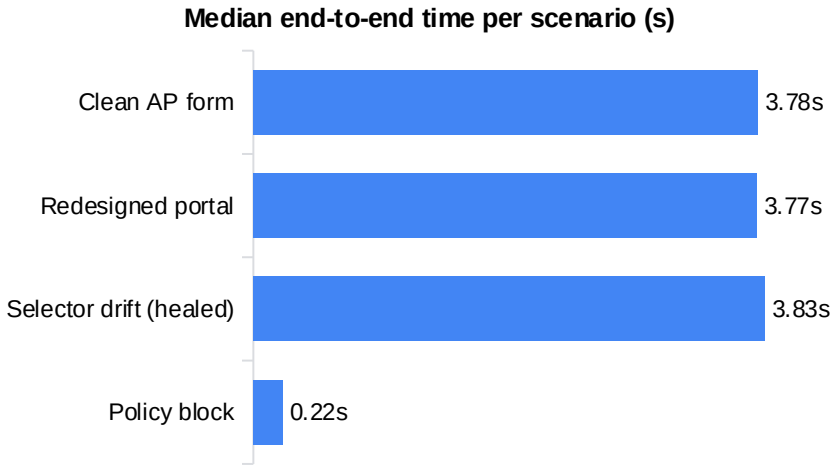
Prototype Performance report/Benchmarking

12/12
benchmark runs as expected

10/10
injection attacks blocked

0
ledger changes from over-limit invoices

855
automated tests passing



Scenario	Result
Clean AP form	3/3 pass, record in ledger once
Redesigned portal	3/3 pass
Selector drift	3/3 - healed, then learned skill
Over-limit invoice	3/3 blocked, 0 ledger changes
Model Armor	10/10 blocked, 0 false positives

scripts/benchmark.py drives the real product against the bundled demo ERP, 3 runs per scenario, and checks the ERP ledger itself; run in offline heuristic mode.

Additional Details/Future Development (if any)

SHIPPED

Today (v9.9.1)

- Five gated steps on Cloud Run
- Traditional and agentic automation
- Mission Control, Model Armor, learned skills

NEXT

Hardening

- Each run as its own Cloud Run job behind a VPC egress firewall
- SSO / OIDC and Identity-Aware Proxy
- Vertex AI embeddings for skill and history search

LATER

Scale-out

- Desktop and API processes beyond the browser
- Scheduled runs and a shared process library
- Agent Platform integration for governed agent fleets

Goal: *every repetitive screen process becomes a governed, testable asset - learned from one demonstration.*

Provide links to your:

GitHub Public Repository

https://github.com/CodeWithMir/Understudy

Demo Video Link (3 Minutes)

https://youtu.be/z9ClL-fcu0k

Final Product Link

https://<cloud-run-service>.run.app

Team JAAAR: Mir Jasimuddin (github.com/CodeWithMir) · Aarya Dutta (github.com/duttaaarya2000)

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

